NextInResearch

Workshop #4: Data Analysis, Statistical Tests, and Applied Computer Science

Key Types of Research Analysis

- **Quantitative**: Numerical data, descriptive & inferential statistics, p-values
- **Qualitative**: Non-numerical (words/audio/video), themes, thematic/content analysis
- **Mixed Methods**: Combines both quantitative and qualitative approaches!

Step 1: Data Preparation

- **Data Cleaning**: Correct errors (Ex: age "290" → "29")
- **Data Transformation**: Standardize formats (Ex: "M", "male" → code 1)
- **Missing Data**: Delete or impute (add an estimate of missing data)

Step 2: Analytical Approach

A. Quantitative Analysis
B. Qualitative Analysis
C. Mixed Methods Integration

Descriptive Statistics:

Summarizes central tendency and spread of data. Usually includes: Mean, Median, Mode, Standard Deviation

Ex: Mean test score 85% and standard deviation of 5 points.

Inferential Statistics

Sample → Population predictions via Regression, T-Test, ANOVA

p-Value & Significance

p-Value < 0.05 = statistically significant

Qualitative Coding

Label text segments. Ex: "unpaid overtime" → [Burnout]

Step 3: Interpretation & Visualization

A. Quantitative Analysis

Thematic Analysis
Group codes into themes. Ex: [Burnout] + [Low Pay] → "Toxic Workplace Culture"

Content Analysis
Count word/phrase frequency. Ex: count "safety" mentions across transcripts

B. Qualitative Analysis

Data Visualization
Charts and plots reveal trends and distributions across datasets and time periods.

Contextualization
Explain significance and limitations. Ex: 40% app engagement spike in December = holiday effect

C. Mixed Methods

Combines quantitative results with qualitative insights for richer, validated conclusions.

Key Reminders

Data must be cleaned and standardized before analysis. Errors and inconsistencies skew results significantly.

Choose analytical approach based on research question type. Quantitative = "how much"; Qualitative = "why/how".

Applied Computer Science

Using Applied Computer Science in your research project:

- **Data analysis**: Visualizing data, coding an app, or automate a process, you can write code to do so.
- **Key skills to learn for your own projects/high school**: Coding in python! Often, introductory courses explain a lot about what computers are and how they work. Python is a universal language and it's a great way to start and once you learn the fundamentals a lot can be built off it.

ChatGPT, Claude, Gemini: Large Language Models (LLMs) are advanced artificial intelligence systems trained on mass amounts of data to understand, summarize, and generate human-like text. By predicting the most statistically probable next words in a sequence, they can answer questions, write code, and solve complex problems